

Heterogeneity in Dynare 7

Dynamics: First-Order Solution and Simulations

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Recursive representation (reminder)

As in Bhandari et al. [2023] (BBEG):

- Aggregate state $Z = [\mathcal{E}, A, \Omega]'$, individual state (θ, a, Z) .
- Individual policies: $\bar{x}(\theta, a, Z)$, $\bar{a}(\theta, a, Z)$.
- Aggregate policies: $\bar{X}(Z)$, $\bar{\Omega}(Z)$.
- Equilibrium equations:

$$\begin{aligned} F(\theta, a, \bar{x}(\theta, a, Z), \mathbb{E}_{\theta'|\theta, \mathcal{E}'}[\bar{x}(\theta', \bar{a}(\theta, a, Z), Z')], \bar{Y}(Z)) &= 0, \\ G(\int \bar{y}(\theta, a, Z) d\Omega(\theta, a), P\bar{X}(Z), \bar{X}(Z), \mathbb{E}_{\mathcal{E}'}\bar{X}(Z'), \mathcal{E}) &= 0, \\ \bar{\Omega}(Z)(B) &= \int \mathbf{1}\{(\theta', \bar{a}(\theta, a, Z)) \in B\} d\pi(\theta' | \theta) d\Omega(\theta, a). \end{aligned}$$

with $\bar{Y}(Z) \subseteq [\mathcal{E}, P\bar{X}(Z), \bar{X}(Z), \mathbb{E}_{\mathcal{E}'}\bar{X}(Z')]$ the relevant aggregate slice.

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First-order approximation

- Recursive law of motion of the aggregate state:

$$\bar{Z}(Z) = [\mathcal{E}, P\bar{X}(Z), \bar{\Omega}(Z)].$$

- Directional derivative along $\hat{Z}$:

$$D_Z \bar{X} \cdot \hat{Z} = \lim_{\alpha \rightarrow 0} \frac{\bar{X}(Z^* + \alpha \hat{Z}) - \bar{X}(Z^*)}{\alpha}.$$

- First-order moving-average representation around Z^* :

$$X_t(\mathcal{E}_t) = X^* + \sum_{s=0}^t \hat{X}_{t-s} \mathcal{E}_s + O(\|\mathcal{E}\|^2),$$

with $\hat{Z}_0 \equiv [1, 0, (\theta, a) \mapsto 0]^\top$, $\hat{Z}_t \equiv D_Z \bar{Z} \cdot \hat{Z}_{t-1}$, $\hat{X}_t \equiv D_Z \bar{X} \cdot \hat{Z}_t$.

Outline

- 1 Differentiate the aggregate equation G
 - relationship between the responses of
 - aggregate variables $\hat{X}_t$,
 - summed heterogeneous variables $(\int \bar{x} d\Omega)_t$,
 - depends on the responses of
 - individual policy functions $\hat{x}_t$,
 - distribution of agents $\hat{\Omega}_t$.
- 2 Differentiate the individual equation F
 - analytical expression linking $\hat{x}_t$ and $\hat{Y}_t$.
- 3 Differentiate the law of motion of the distribution
 - analytical expression linking $\hat{\Omega}_t$ and $\hat{x}_t$.
- 4 Stack into a sparse linear system in $\{\hat{X}_t\}_{t \geq 0}$.

Step 1 (aggregate block): differentiate G

- We want $\hat{X}_t$, hence $\hat{Z}_t = [\mathcal{E}_t, P\hat{X}_{t-1}, \hat{\Omega}_t]$, with $\hat{\Omega}_t = D_Z \bar{\Omega} \cdot \hat{Z}_t$.
- First derivative of G in the direction $\hat{Z}_t$:

$$D_Y G \cdot \hat{Y}_t + D_{\bar{x}} G \cdot D_Z \left(\int \bar{x} d\Omega \right) \cdot \hat{Z}_t = 0,$$

where $\hat{Y}_t = [\mathcal{E}_t, P\hat{X}_{t-1}, \hat{X}_t, \hat{X}_{t+1}]$.

- Product-rule split of the aggregation:

$$D_Z \left(\int \bar{x} d\Omega \right) \cdot \hat{Z}_t = \int \underbrace{D_Z \bar{x} \cdot \hat{Z}_t}_{\equiv \hat{\bar{x}}_t} d\Omega^* + \int \bar{x}^* d\hat{\Omega}_t.$$

Step 2 (individual block): differentiate F via the backward operator

- Recall the heterogeneous functional from Session 2:

$$\mathcal{F}[\bar{x}, \bar{x}^+, Y](\theta, a) \equiv F(\theta, a, \bar{x}(\theta, a), \int \bar{x}^+(\theta', p \bar{x}(\theta, a)) d\pi(\theta' | \theta), Y).$$

- Recall the IFT-defined backward operator $\mathfrak{x}(\bar{x}^+, Y)$ from Session 2:

$$\mathcal{F}[\mathfrak{x}(\bar{x}^+, Y), \bar{x}^+, Y] = 0.$$

- Fréchet derivatives of $\mathfrak{x}$ at $(\bar{x}^*, Y^*)$ in the directions $\hat{Z}_t, t \geq 0$:

$$\begin{aligned} x_0 &= D_Y \mathfrak{x} = -(D_x \mathcal{F})^{-1} D_Y \mathcal{F}, \\ x_{s+1} &= D_{x^+} \mathfrak{x} \cdot x_s = -(D_x \mathcal{F})^{-1} D_{x^+} \mathcal{F} \cdot x_s. \end{aligned}$$

- Anticipated response of the individual policy:

$$\hat{x}_t(\theta, a) = \sum_{s \geq 0} x_s(\theta, a) \hat{Y}_{t+s}.$$

Step 3 (distribution): differentiate the law of motion

- Differentiate the law of motion of the distribution in direction $\hat{Z}_t$:

$$\frac{d\hat{\Omega}_{t+1}}{d\hat{a}'d\hat{\theta}'} = \nabla_{a'} \cdot \hat{\omega}_{t+1},$$

$$\hat{\omega}_{t+1} = \mathcal{L}^{(a)} \hat{\omega}_t - \mathcal{M} \hat{a}_t, \quad \hat{\omega}_0 = 0.$$

- With:

$$(\mathcal{L}^{(a)} y)(\theta', a') = \int \bar{\Lambda}(a', \theta, a) y(\theta, a) d\Omega^*(\theta, a) d\pi(\theta'|\theta),$$

$$(\mathcal{M} y)(\theta', a') = \int \int \bar{\Lambda}(a', a, \theta) D_a \bar{a}^*(\theta, a) y(a) d\Omega^*(\theta, a) d\pi(\theta'|\theta),$$

$$\bar{\Lambda}(a', a, \theta) = \prod_{i=1}^{n_a} \delta(a'^i - \bar{a}^{*,i}(\theta, a)), \quad n_a \text{ the number of individual states.}$$

Putting it together

- Substituting Steps 1–3 into the aggregate block, for all t :

$$D_Y G \cdot \hat{Y}_t + D_{\mathbf{x}} G \cdot \sum_{s \geq 0} J_{t,s} \hat{Y}_s = 0,$$

where

$$J_{0,s} = \int \mathbf{x}_s(\theta, \mathbf{a}) d\Omega^*$$

$$J_{t,s} = J_{t-1,s-1} + \mathcal{J}^{(\mathbf{a})} \cdot \left(\mathcal{L}^{(\mathbf{a})} \right)^{t-1} \mathcal{M} \cdot p \mathbf{x}_s,$$

$$\mathcal{J}^{(\mathbf{a})} \cdot y = \int D_{\mathbf{a}} \bar{\mathbf{x}}^*(\theta, \mathbf{a}) y(\theta, \mathbf{a}) d\mathbf{a}$$

- Truncate to a finite horizon $T \rightarrow$ sparse linear system in $\{\hat{X}_t\}_{t=0}^{T-1}$.

The fake-news algorithm: the sequence-space viewpoint

Auclert et al. [2021] (ABRS): Suppose the system rests at the steady state from period $T + 1$ onward, and at period $t \leq T$ households learn of a fake shock $\hat{Y}$ to land at period T . Denote $\hat{x}_t$ the period- t policy response.

- Period T (the shock lands): $\hat{x}_T = D_Y \mathfrak{F} \cdot \hat{Y} = -(D_x \mathcal{F})^{-1} D_Y \mathcal{F} \cdot \hat{Y} \equiv x_0 \cdot \hat{Y}$.

Traditional Fréchet derivative w.r.t. Y .

- Period $t < T$ (rolled backward): $\hat{x}_t = D_{x^+} \mathfrak{F} \cdot \hat{x}_{t+1} = -(D_x \mathcal{F})^{-1} D_{x^+} \mathcal{F} \cdot \hat{x}_{t+1} \equiv x_{T-t} \cdot \hat{Y}$.

Fréchet derivative w.r.t. $\bar{x}^+$, propagated backward $T - t$ times.

→ the IFT operator $\mathfrak{F}$ in (BBEG) and the user-coded backward step in (ABRS) generate identical x_s .

→ One can prove that (BBEG) and (ABRS) are asymptotically equivalent

What's in Dynare?

(BBEG) is the functional-analysis counterpart of (ABRS):

- same fundamental objects;
- state space instead of sequence space;
- (BBEG) naturally handles kinks in the policy functions;
- (BBEG) allows X_{t-1} in the heterogeneous block.

→ take the best of both worlds in Dynare:

- general character of (BBEG),
- speed of (ABRS),
- more than two individual states.

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heterogeneity_solve: syntax and options

Syntax:

```
heterogeneity_solve ;  
heterogeneity_solve (OPTIONS...);
```

Prerequisite. A steady state must already be in `oo_.heterogeneity`
→ call to `heterogeneity_load_steady_state` or
`heterogeneity_compute_steady_state`.

Options.

- `truncation_horizon = INTEGER`: time horizon T for the sequence-space Jacobians.
Default: 300.

Side effects. Populates `oo_.heterogeneity.dr`. The user-facing subfield is `dr.G`; the rest are internal and may move.

heterogeneity_solve: documented output

The general-equilibrium Jacobian `oo_.heterogeneity.dr.G.(VAR).(SHOCK)` is a $T \times T$ matrix.

- Entry (t, s) : linear response of VAR in period t to a unit deviation of SHOCK in period s , treated as known at time 0
- The impulse response is the first column

```
oo_.heterogeneity.dr.G.K.Z(:, 1)
```

- Anticipated sequence of shocks `SHOCK_PATH`
→ Simple matrix multiplication with G

$$\underbrace{\text{SIM}_{\text{VAR}}}_{T \times 1} = \underbrace{\text{dr.G.VAR.SHOCK}}_{T \times T} \cdot \underbrace{\text{SHOCK_PATH}}_{T \times 1}.$$

- Used internally by `heterogeneity_simulate`

Beneath `dr.g`: undocumented outputs I

Some useful intermediate quantities live on `oo_.heterogeneity.dr`.

- *Field names and layout are not part of the stable interface!*
- Decomposition of the response of $\int \bar{x} d\Omega$ to anticipated shocks

Sub-fields:

- `dr.J_ha.(x).(Y) [T×T]`:
Entry (t, s) is the total response of $\int \bar{x} d\Omega$ at t to a unit aggregate anticipated shock on Y at s .
→ Sequence-space Jacobian of summed heterogeneous variables
- `dr.F.(x).(Y) [T×T]`:
Column s is the impulse response to a shock on Z announced at date 0 to happen on period s , and retracted at date 1.
→ Useful to diagnose at which horizon a channel kicks in.

Beneath `dr.g`: undocumented outputs II

- `dr.curlyYs.(Y).(x) [T×1]:(∫  $\times_s d\Omega^*$ )s≥0`
 → Time-0 response to shocks announced at date 0 to happen at date s
- `dr.curlyDs.(Y) [T×shock dims×state dims]:`
 → Shift of the histogram after a shock announced at date 0 to happen at date s
 → How much the distribution drifts per anticipation lag

Beneath `dr.G`: the aggregate-block synthetic writing

- `dr.J.(eq).(var)`: Jacobian of the aggregate residuals function G
 - Stored equation-by-equation as `[timing, value]` pairs.
 - `timing` $\in \{-1, 0, +1\}$: lag, current/exo, lead.
→ Same convention as standard Dynare's `dr.ghx/dr.ghu`.
 - `value` is the scalar partial derivative.
- Unrolling `dr.J` to dense $T \times T$ block-Toeplitz form makes the general-equilibrium closure explicit, as in (ABRS):

$$\text{dr.G} = -\left(\text{dr.J}_Y + \text{dr.J}_x \cdot \text{dr.J}_{\text{ha}}\right)^{-1} \cdot \text{dr.J}_g$$

Channel decomposition

The full general-equilibrium response $\text{dr.G.}(\text{var}).(\text{shock})$ hides *how* the shock propagates.

→ With a flat DAG, only 2 blocks: heterogeneous and aggregate:

$$\text{dr.G.X.}\mathcal{E} = \sum_{\text{var} \in Y} \text{dr.J_ha.X.var} \cdot \text{dr.G.var.}\mathcal{E}.$$

Each term is the heterogeneous-block channel through aggregate *var*.

→ Reveal which variables drive summed heterogeneous variables

Channel decomposition

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→ Reveal which variables drive summed heterogeneous variables

Example: Krusell-Smith, TFP shock $\mathcal{E}_Z$, aggregate capital K :

$$\text{dr.G.K.}\mathcal{E}_Z = \underbrace{\text{dr.J_ha.K.r} \cdot \text{dr.G.r.}\mathcal{E}_Z}_{\text{interest-rate channel}} + \underbrace{\text{dr.J_ha.K.w} \cdot \text{dr.G.w.}\mathcal{E}_Z}_{\text{wage channel}}.$$

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heterogeneity_simulate: prerequisites and syntax

Prerequisite. `heterogeneity_solve` must have run first, so that `oo_.heterogeneity.dr.G` is in memory.

Syntax:

```
heterogeneity_simulate ;  
heterogeneity_simulate (OPTIONS...) [VARIABLE_NAME...];
```

Auto-detected mode.

- Stochastic simulation by default;
- News-shock sequence as soon as the `shocks` block carries `periods/values` clauses.

heterogeneity_simulate: stochastic simulation I

Innovations drawn from $\mathcal{N}(0, M_Sigma_e)$.

- Always returns IRFs to each declared shock (1 std-dev impulse).
- Additionally returns a stochastic sample path if `periods > 0`.
- The shock process equation must be in the `model` block
→ Dynare simulates innovations, not the exogenous variable itself, unlike (ABRS)

heterogeneity_simulate: stochastic simulation II

Options (forbidden in news-shock mode):

- `irf` = INTEGER: IRF horizon. Default: 40.
- `periods` = INTEGER: simulation length; 0 means IRFs only. Default: 0.
- `irf_shocks` = (SHOCK_NAME ...): restrict the IRF set. Default: all aggregate shocks.
- `relative_irf`: IRFs as percent deviation from a 1-unit shock.

Outputs

- `oo_.irfs.(var)_(shock)`: $T \times 1$ IRF for each variable–shock pair (always populated).
- `oo_.endo_simul`: $n_{\text{vars}} \times (\text{periods} + 1)$ simulated paths in levels (only when `periods > 0`).

heterogeneity_simulate: news-shock sequence

Perfect-foresight responses to anticipated shocks. Triggered by:

```
shocks;  
    var Z;  
    periods 1:5;  
    values 0.01, 0.008, 0.006, 0.004, 0.002;  
end;
```

Households learn the full sequence at $t = 0$.

→ Mutually exclusive with the stochastic-mode options below.

Outputs:

- `oo_.endo_simul`: simulated paths in levels.
- `oo_.exo_simul`: realised news-shock paths.

heterogeneity_simulate: Common comments

Plotting and IO (both modes):

- `nograph`, `nodisplay`, `graph_format = FORMAT, tex`.
- `irf_plot_threshold = DOUBLE, print, noprint`.

Outputs:

- Stored under standard Dynare slots, not under `oo_.heterogeneity`.
- Internally, both modes contract well-chosen shock paths with `oo_.heterogeneity.dr.G.(var).(shock)` through matrix-vector multiplication

Exogenous processes in Dynare

A frequent surprise when porting an sequence-space-Jacobian notebook to Dynare:

- **Sequence-space Jacobian** consumes the *full sequence of aggregate-shock values* $\{\hat{X}_t^{(Z)}\}_{t=0}^{T-1}$ directly. It is user-supplied!
- **Dynare** uses the exogenous-process equation, e.g.

```
varexo eps_Z;
parameters rho_Z;
model;
    Z = rho_Z * Z(-1) + eps_Z;
    /* ... */
end;
shocks;
    var eps_Z; stderr 0.01;
end;
```

→ Dynare simulates *innovations* $\mathcal{E} = (\mathcal{E}_t)_{t \geq 0}$. The IRF of any variable is then $\text{oo_heterogeneity.dr.G.}(\text{var}).\text{eps_Z} \cdot \mathcal{E}$.

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Backing out heterogeneous responses

heterogeneity_simulate reports only aggregate paths!

→ A bit of work is necessary to retrieve the responses of policy functions and distribution

- **Policy response:** Given the equilibrium aggregate path $\{\hat{Y}_t\}_t$

$$\hat{x}_t(\theta, a) = \sum_{s \geq 0} x_s(\theta, a) \hat{Y}_{t+s}.$$

→ $\{\hat{Y}_t\}_t$ can be found in the first column of `dr.G` subfields

- **Histogram response:** fed by `dr.curlyDs` (the path $\{\hat{Y}_t\}$ again from `dr.G`).

$$\begin{aligned} \hat{\omega}_0 &= 0, \\ \hat{\omega}_{t+1} &= \bar{\Lambda} \hat{\omega}_t + \sum_{s \geq 0} \text{dr.curlyDs}_s \cdot \hat{Y}_{t+s}. \end{aligned}$$

where $\bar{\Lambda}$ is the steady-state push-forward operator

→ Workshop .m helpers in `exercises/: het_policy_responses.m`,
`het_distribution_responses.m`

Animated cross-section: time-lapse responses

Examples (Krusell–Smith, TFP shock).

- Consumption-policy response on the joint (θ, a) grid
- Marginal wealth distribution response
- Joint distribution response over (θ, a)

► Policy

► Aggregated

► Disaggregated

Bibliography

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